KAVYA GUPTA

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PROFILE

Computer Science undergraduate with hands-on experience building real-time computer vision systems (OpenCV, YOLOv8, MediaPipe) and deep learning pipelines (PyTorch, Scikit-Learn). Interested in applying computer vision to biomedical and healthcare imaging problems.

EDUCATION

Vellore Institute of Technology, Bhopal 2023 – 2027
B.Tech in Computer Science and Engineering — CGPA: 8.67 Bhopal, Madhya Pradesh

TECHNICAL SKILLS

Computer Vision: OpenCV, YOLOv8 (Object Detection), MediaPipe (Facial Landmark Detection), Image & Video Processing, Real-time Detection Pipelines

AI/ML & Deep Learning: PyTorch, Scikit-Learn, Machine Learning, Deep Learning, Generative AI, LLMs, RAG, NLP, Explainable AI (SHAP)

Languages: Python, C++, JavaScript, SQL

Frameworks: FastAPI, Flask, Streamlit, Node.js, Express.js, REST APIs

Databases: MongoDB, PostgreSQL, SQLite, Redis

Tools/Cloud: Git, GitHub, Postman, VS Code, Linux, Azure, OCI

Coursework: DSA, OOP, DBMS, OS, Computer Networks

PROJECTS

AI Driver Monitoring System (Computer Vision) | *Python, OpenCV, MediaPipe, YOLOv8*

- Built a real-time computer vision pipeline for driver fatigue and distraction detection from live webcam video, fusing facial-landmark, head-pose, and object-detection signals into a single risk assessment.
- Implemented drowsiness and yawning detection using Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) algorithms computed from MediaPipe facial landmarks.
- Designed head-pose estimation for distraction detection and integrated a YOLOv8 object-detection model to identify mobile-phone usage in real time.
- Engineered a multi-factor fatigue-scoring algorithm and Low/Medium/High risk classification system, with real-time voice alerts triggered on high-risk states.

AI Proposal Evaluation System (R&D) | *Python, FastAPI, Scikit-Learn, SHAP, Generative AI*

- Built a full-stack AI proposal evaluation platform (FastAPI backend, Streamlit frontend) with REST APIs for document parsing, automated evaluation, and AI-assisted report generation.
- Designed a modular ML evaluation pipeline combining feature extraction, financial validation, ML scoring, and explainable AI, achieving 91.3% prediction accuracy.
- Integrated SHAP-based model explainability and automated PDF report generation, delivering transparent evaluations in under 5 seconds.

Safe.AI | *HTML, CSS, JS, Redis, MongoDB, AI Moderation APIs*

- Developed a real-time multi-modal (text + image) content moderation engine analyzing 10,000+ samples for toxicity and unsafe content.
- Achieved 95%+ detection accuracy by integrating Bytez.js and Sightengine APIs for image and text classification.
- Reduced moderation latency by 35% using Redis-based asynchronous processing, with scalable MongoDB storage for 50,000+ logs.

CERTIFICATIONS

- Generative AI Using IBM Watsonx (IBM)
- Oracle Cloud Infrastructure 2025 Certified Data Science Professional
- Microsoft Azure Data Fundamentals (DP-900)

ACHIEVEMENTS

- Machine Learning Lead, Google Developers Group Club (GDGC), Oct 2025 – Sep 2026
- SIH Hackathon 2025 Finalist — Top 45 of 950 teams (Internal Round)
- 2nd Runner-up, CyberShield Hackathon (WiCyS Club)